POWER BI PROJECT REPORT

Restaurant & Food Delivery Analytics Dashboard

An interactive 6-page Business Intelligence dashboard

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Tool Used	Microsoft Power BI Desktop (Power Query, DAX, Data Modeling)
Project Type	Business Intelligence / Data Visualization
Dashboard Pages	6 (Executive, Customer, Restaurant, Delivery, Advanced, Drill-through)

200K	20K	56K	2023-25
Total Orders	Customers	Restaurants	Period

Prepared using Microsoft Power BI Desktop

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About this report

- This report documents the design, data preparation, modelling and findings of a 6-page interactive Power BI dashboard built for restaurant and food-delivery analytics.
- All figures quoted are taken from the dashboard screenshots included in Section 7.

1. Project Title

Restaurant & Food Delivery Analytics Dashboard - an end-to-end Power BI report that analyses revenue, customers, restaurant quality and delivery operations for restaurants across Bengaluru localities.

165M Total Revenue	**200K** Total Orders	**824.57** Avg Order Value	**44.55 min** Avg Delivery Time
50.28% Repeat Customers	**3.70** Avg Rating	**80.08%** Orders Delivered	**9.98%** Cancelled

2. Dataset Description

The project combines restaurant listing data with customer order data. It covers about 200K orders, 20K customers and 56K restaurant records across multiple Bengaluru localities (Banashankari, BTM, Koramangala, Indiranagar, Whitefield, Church Street, etc.) for the years 2023 - 2025.

Category	Key Columns
Restaurant	name, PrimaryCuisine, rate (rating), votes, Cost_Bucket (Budget / Mid Range / Premium), ServiceType (Cafes, Desserts, Delivery, Dine-out, Drinks & nightlife, Pubs and bars, Buffet), Online Order, Table Booking
Customer	CustomerID, CustomerType (New / Repeat), Signup Date / Year
Order	OrderID, Order Date, Revenue, Order Status (Delivered / Delayed / Cancelled), Delivery Status (On Time / Late), Delivery Time (minutes)
Location	FilterCity (locality / area)

3. Objectives

- Build a single executive view of revenue, orders, average order value and delivery time.
- Understand customer behaviour - new vs repeat customers, orders per customer and top-revenue customers.
- Evaluate restaurant performance by cuisine, rating, cost bucket and revenue.
- Monitor delivery and operations - delays, late orders, cancellations and city-wise delivery time.
- Provide interactive exploration through slicers, drill-through, a KPI selector and Q&A.
- Convert raw data into actionable business insights.

4. Data Cleaning Performed

Data preparation was carried out in the Power Query Editor before modelling:

- Removed duplicate records and blank / null rows.
- Standardised column data types (dates, whole numbers, decimals, text).

- Cleaned the rating column (removed text such as "/5", "NEW", "-") and converted it to a numeric *rate* field.
- Trimmed and standardised text values (restaurant names, cuisines, city / locality names).
- Extracted the primary cuisine from multi-cuisine strings into *PrimaryCuisine*.
- Created the *Cost_Bucket* column (Budget / Mid Range / Premium) from the approximate cost-for-two field.
- Created a *FilterCity* column to give clean locality names for slicers and charts.
- Derived *CustomerType*, *Year* and delivery status (On Time / Late) fields.
- Handled missing values (replaced with defaults or filtered out where unusable).
- Renamed columns and tables with business-friendly names.

Note: a few records show values such as ('Rated 4.0' in the restaurant name field (visible in the Top 10 Restaurants table). This is a parsing artefact from the source data and is listed under Future Improvements.

5. Data Model

A **star-schema** model was used: a fact table at the centre, descriptive dimension tables around it, joined by one-to-many relationships with single-direction filtering from each dimension to the fact table.

Table	Type	Role / Key
Orders	Fact	Revenue, order status, delivery time
Customers	Dimension	CustomerID, CustomerType
Restaurants	Dimension	Restaurant name, cuisine, rating, cost bucket
Location	Dimension	FilterCity
Date Table	Dimension	Date, Month, Year
KPI Name	Helper	Field parameter for the KPI selector

6. DAX Measures

Core measures used across the dashboard:

Core KPIs

```
Total Revenue = SUM(Orders[Revenue])
Total Orders = COUNTROWS(Orders)
Avg Order Value = DIVIDE([Total Revenue], [Total Orders])
Avg Delivery Time = AVERAGE(Orders[DeliveryTime])
```

Customer Metrics

```
Total Customers = DISTINCTCOUNT(Orders[CustomerID])
Repeat Customers =
    CALCULATE(DISTINCTCOUNT(Orders[CustomerID]),
              Customers[CustomerType] = "Repeat")
Repeat Customer % = DIVIDE([Repeat Customers], [Total Customers])
Orders Per Customer = DIVIDE([Total Orders], [Total Customers])
```

Restaurant Metrics

```
Total Restaurants = DISTINCTCOUNT(Restaurants[name])
Average Rating = AVERAGE(Restaurants[rate])
Average Votes = AVERAGE(Restaurants[votes])
Revenue per Restaurant = DIVIDE([Total Revenue], [Total Restaurants])
```

Delivery & Operations

```
Late Orders = CALCULATE([Total Orders], Orders[DeliveryStatus] = "Late")
Cancelled Orders = CALCULATE([Total Orders], Orders[OrderStatus] = "Cancelled")
Late Order % = DIVIDE([Late Orders], [Total Orders])
Cancellation Rate = DIVIDE([Cancelled Orders], [Total Orders])
```

The KPI selector on the Advanced Insights page uses a **field parameter** (*KPI Name*) so the user can switch the charts between Average Delivery Time, Orders, Average Order Value and Revenue.

7. Dashboard Screenshots

The report has six pages, designed with a consistent red-and-white theme: KPI cards at the top and detailed visuals beneath. Each page below is followed by the main insights it reveals.

Page 1 - Executive Overview

High-level KPIs, revenue by city, order-status breakdown, Top 10 restaurants, revenue / order trends and a city-wise filter.

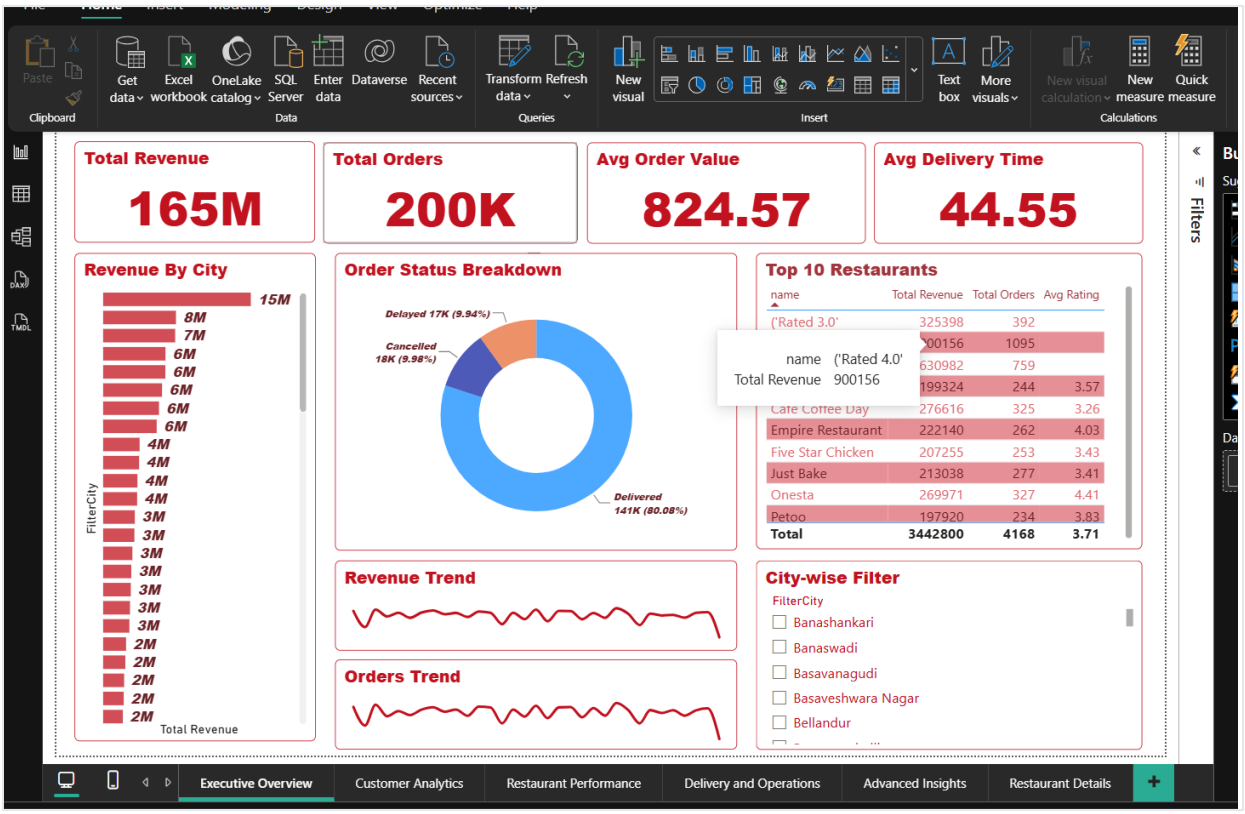


Figure 1: Page 1 - Executive Overview

Key insights

- 165M revenue from 200K orders; average order value 824.57
- 80.08% delivered, 9.94% delayed, 9.98% cancelled
- Top city contributes about 15M revenue, almost double the next

Page 2 - Customer Analytics

Total / repeat customers, repeat %, orders per customer, orders by customer type, new vs repeat split, top-10 customers and table-booking slicer.

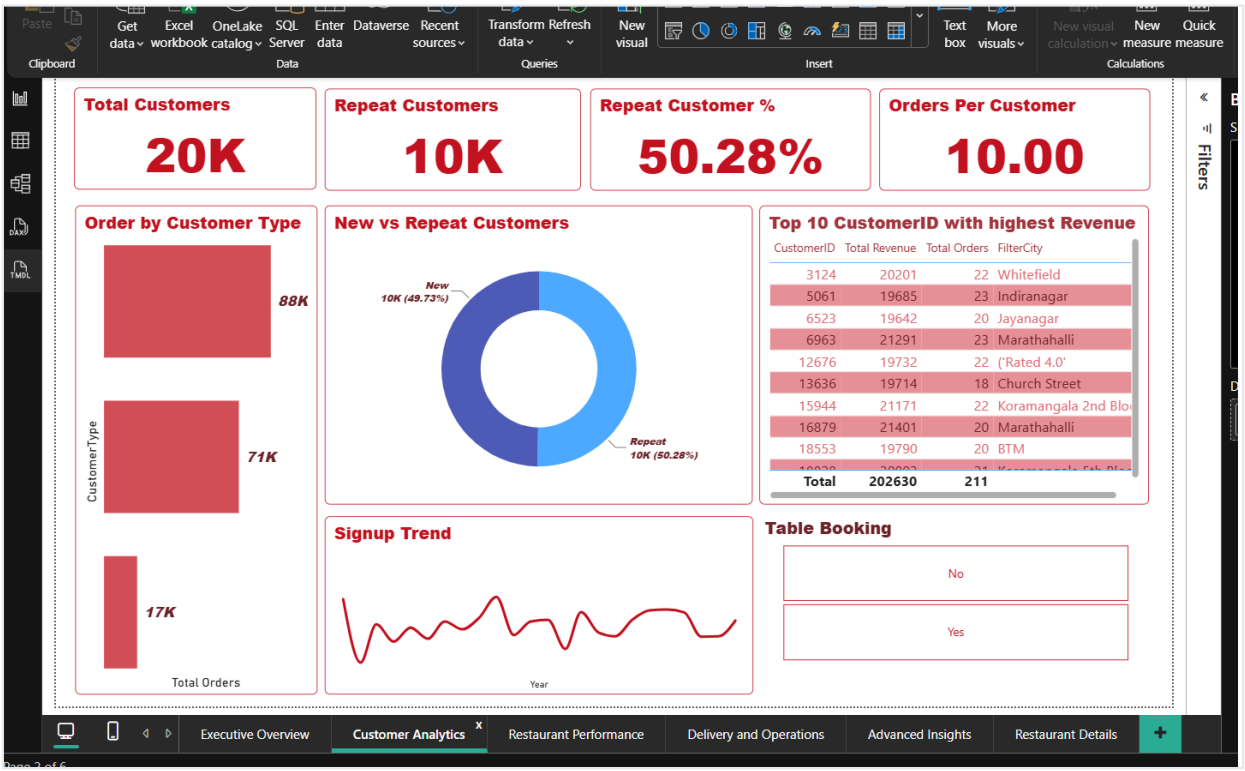


Figure 2: Page 2 - Customer Analytics

Key insights

- 20K customers, of which 10K (50.28%) are repeat customers
- Orders per customer average 10.00 - strong engagement
- Top-10 customers each generate roughly 19K-21K revenue

Page 3 - Restaurant Performance

Restaurant KPIs, orders by cuisine, rating-wise revenue, online-order slicer, rating by cuisine, cost-bucket split and revenue by top 10 restaurants.

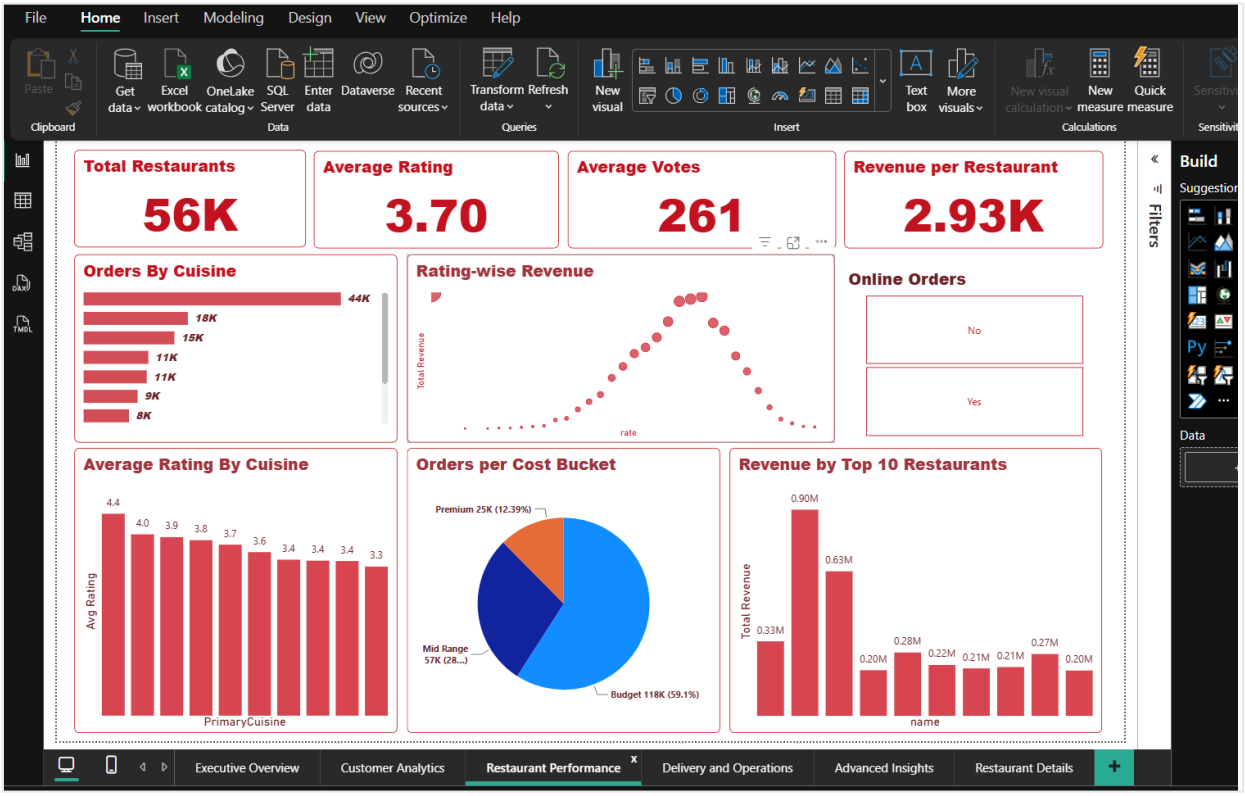


Figure 3: Page 3 - Restaurant Performance

Key insights

- Average rating 3.70 and 261 votes; revenue per restaurant 2.93K
- Budget restaurants drive 59.1% of orders, Mid Range 28.5%, Premium 12.4%
- Revenue peaks for restaurants rated around 3.8-4.2

Page 4 - Delivery and Operations

Delivery time, late and cancelled orders, city-wise delivery time and cancellations, order value by service type, delivery details and status breakdown.

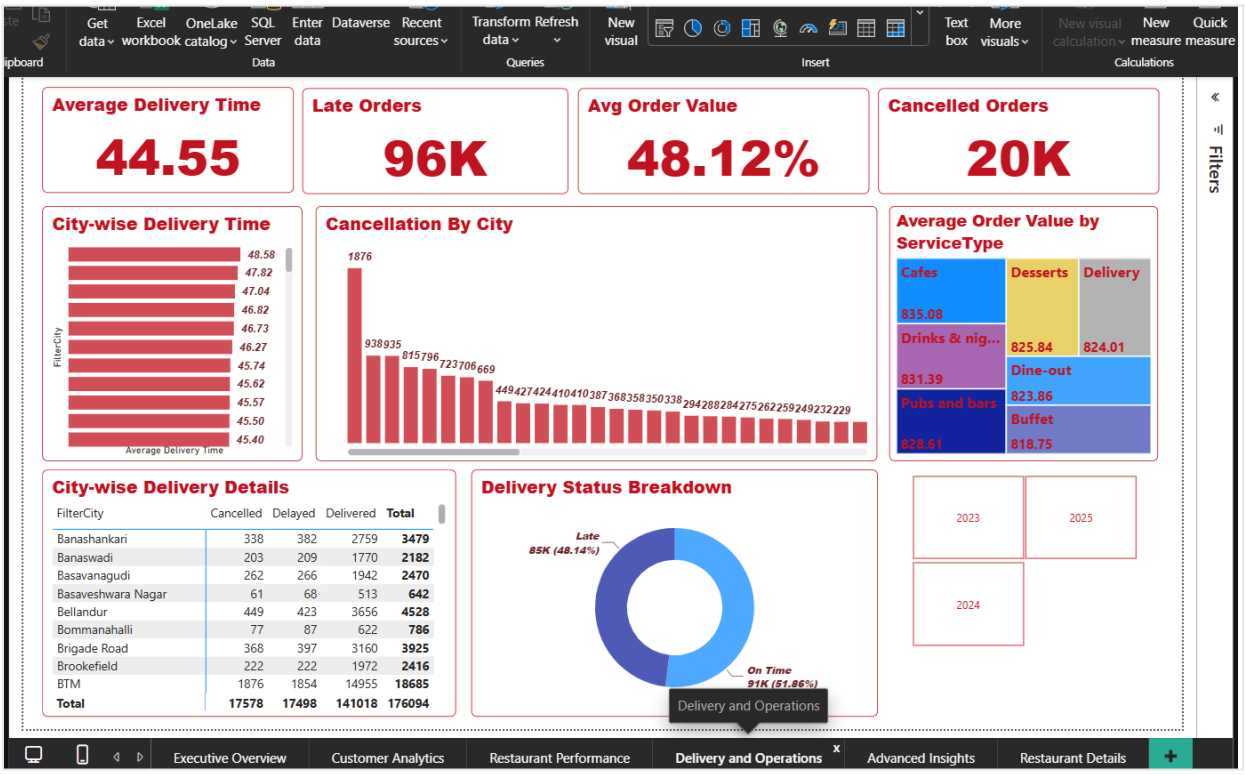


Figure 4: Page 4 - Delivery and Operations

Key insights

- Average delivery time is 44.55 min; city averages range 45-48.6 min
- BTM has the most cancellations (1,876), roughly double the next area
- Cafes have the highest average order value (835.08)

Page 5 - Advanced Insights

A KPI selector (field parameter) that changes the trend chart and the rating-vs-KPI scatter plot by cost bucket, plus a Q&A visual.



Figure 5: Page 5 - Advanced Insights

Key insights

- Monthly revenue stays between 4.6M and 5.3M
- Budget restaurants peak at mid-range ratings; Premium shows larger bubbles at higher ratings
- Q&A allows natural-language questions such as "Top 5 Restaurants by rate"

Page 6 - Restaurant Details (Drill-through)

A drill-through page showing KPIs, customer-type orders, delivery status and trends for a selected restaurant (example: Cafe Coffee Day).

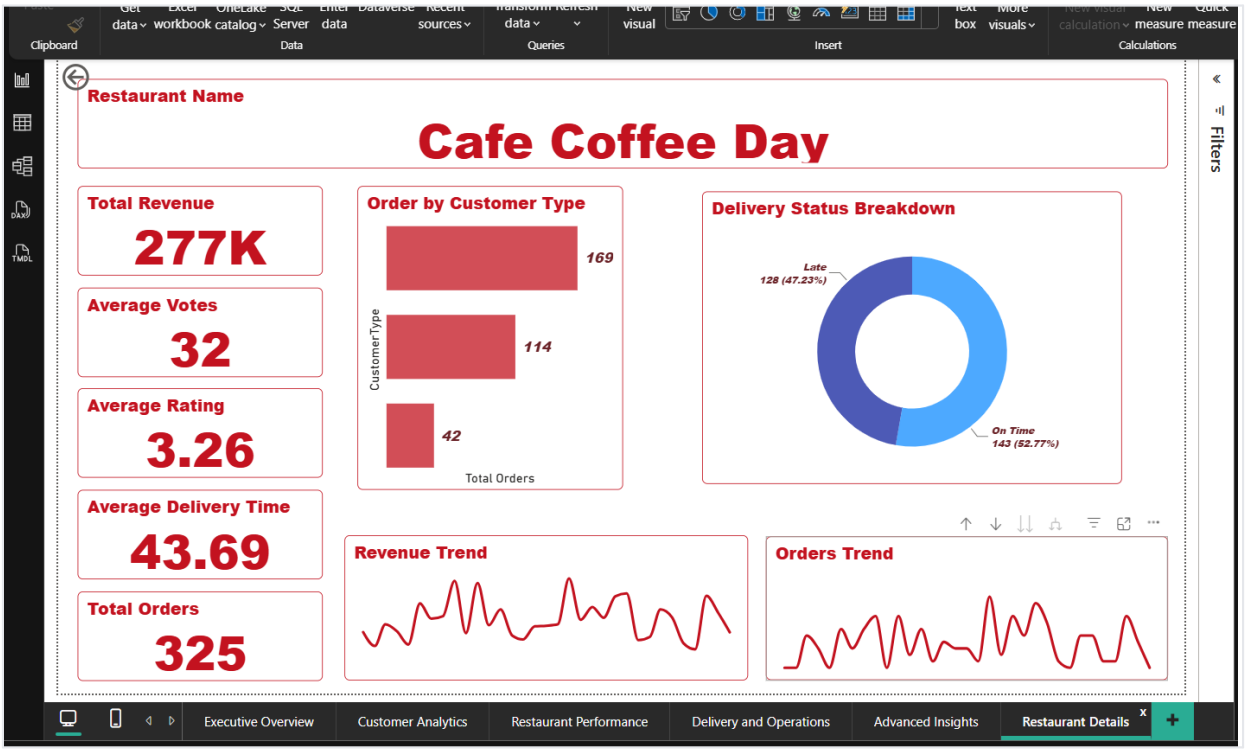


Figure 6: Page 6 - Restaurant Details (Drill-through)

Key insights

- Cafe Coffee Day: 277K revenue, 325 orders, rating 3.26
- Average delivery time 43.69 min with 47.23% late deliveries
- Users can drill through from any restaurant on other pages

8. Key Findings

Business Performance

- The business generated **165M revenue from 200K orders**, with an average order value of about 824.57.
- Revenue is concentrated in a few localities - the top city contributes about 15M, nearly double the next one (about 8M).
- The monthly revenue trend stays stable in the 4.6M - 5.3M range, with a sharp drop in the final period (4.26M), likely due to incomplete data for that period.

Orders & Delivery

- **80.08%** of orders are delivered, while 9.94% are delayed and 9.98% are cancelled.
- Average delivery time is 44.55 minutes; city-wise averages range from about 45 to 48.6 minutes.
- **BTM has the highest number of cancellations (1,876)** - roughly double the next locality - making it the top area for operational improvement.
- Roughly half of deliveries arrive late (about 48%), indicating a clear SLA / logistics gap.

Customers

- There are 20K customers and **50.28% are repeat customers**, with a nearly even new-vs-repeat split.
- Each customer places 10 orders on average, which indicates strong engagement.
- Top-10 customers each generate roughly 19K-21K in revenue with 18-23 orders.

Restaurants & Cuisine

- Average restaurant rating is 3.70 with an average of 261 votes.
- The top cuisine accounts for 44K orders, far ahead of the next (about 18K).
- Budget restaurants drive 59.1% of orders, Mid Range 28.5% and Premium 12.4%.
- Revenue peaks for restaurants rated around 3.8-4.2; very low and very high ratings generate less revenue.
- Average order value is highest for Cafes (835.08) and lowest for Buffet (818.75) - a narrow range across service types.

Example - Cafe Coffee Day

- 277K revenue, 325 orders, rating 3.26, average delivery time 43.69 min and about 47% late deliveries - a high-volume brand with below-average rating and delivery reliability.

9. Conclusion

This project shows how Power BI can turn raw restaurant and order data into a complete decision-support tool. The six-page dashboard covers executive KPIs, customer behaviour, restaurant quality, delivery operations, advanced KPI exploration and restaurant-level drill-through.

The analysis shows that the business is **revenue-stable with strong customer loyalty (50% repeat rate)** but faces **operational challenges** - nearly half of deliveries are late and about 10% of orders are cancelled, particularly in high-volume localities like BTM. Focusing on delivery logistics,

improving ratings of high-volume restaurants and promoting mid-range / premium options can increase both customer satisfaction and revenue.

Future Improvements

- Fix parsing errors in the restaurant name field (values like ('Rated 4.0').
- Verify the label of the Avg Order Value card on the Delivery page (it shows a percentage) and align late-order counts between cards and charts.
- Add forecasting for revenue and orders.
- Add peak-hour / day-of-week analysis and customer segmentation (RFM).
- Publish to Power BI Service with scheduled refresh and row-level security.